COMMUNITY EMERGENCY RESPONSE APP (CERA)

Applied Research Project

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COMMUNITY EMERGENCY RESPONSE APP (CERA) – PROJECT PROPOSAL

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INTRODUCTION

TOPIC OVERVIEW

Emergencies like natural disasters, medical situations, or sudden infrastructure failures often create panic and confusion. In these moments, quick and reliable communication between residents, volunteers, and local coordinators is critical. However, many communities still don't have a single, easy-to-use platform that brings everyone together during emergencies. Instead, people often rely on scattered channels like phone calls, text messages, or social media, which may not reach the right people at the right time.

With smartphones being widely available, a mobile-first app can make a meaningful difference. Our project, the **Community Emergency Response App (CERA)**, is designed to give people a simple and reliable way to report emergencies, allow volunteers to respond faster, and enable coordinators to manage situations in real time. Unlike social media or fragmented tools, CERA provides a unified platform focused solely on emergency reporting, response, and coordination.

FRAMING THE PROBLEM

The central problem we are addressing is the lack of a structured, community-driven tool for emergency reporting and volunteer coordination. When an emergency occurs, people often rely on word of mouth, scattered social media updates, or delayed official announcements. These channels are unreliable because the information may not reach the right people at the right time, and in many cases, it lacks accuracy or proper follow-up.

As a result, two major challenges arise. First, response times are often delayed because volunteers or coordinators are not notified immediately when an incident happens. Even when information circulates, it is not directed to those who can provide help in that moment. Second, resources are poorly allocated due to the absence of a structured system. Volunteers may respond unevenly—some incidents may attract too many responders while others are neglected entirely—simply because there is no platform to organize efforts based on location, urgency, and availability.

Our project focuses on three central research and design questions:

How can a mobile app help residents report emergencies quickly and reliably?
→ We aim to design a simple reporting process where a resident can log an emergency in just a few taps, including location and a brief description.

What is the best way to connect available volunteers with nearby incidents?
→ CERA will send real-time notifications to volunteers in the affected area, allowing them to accept tasks instantly.

Can features like real-time location and push notifications improve response time without overwhelming users?
→ We will experiment with concise, actionable notifications to ensure clarity and reduce notification fatigue.

These questions matter because faster communication and clearer task allocation can prevent confusion, minimize damage, and even save lives in high-stakes situations

PROPOSED SOLUTION

The Community Emergency Response App (CERA) is designed to function as a three-way communication bridge between residents, volunteers, and coordinators. Its main goal is to streamline emergency reporting, volunteer response, and incident management in a single, mobile-first platform.

Emergency Reporting
CERA allows residents to quickly report emergencies such as fires, floods, accidents, or medical situations. The reporting process is simple, requiring only the type of emergency, a short description, and the location, which is automatically detected using GPS. To increase clarity and credibility, residents can also attach photos or short videos, ensuring that coordinators and volunteers receive accurate and actionable information.

Volunteer Matching System
Once an incident is reported, CERA notifies nearby volunteers through push notifications. Volunteers can accept or decline tasks based on their availability and proximity, ensuring that only those who are free and nearby are dispatched. This approach reduces delays and prevents confusion or duplication of effort.

Coordinator Dashboard
Coordinators have access to a structured dashboard that displays all reported incidents. Through this interface, they can track which emergencies are active, assign volunteers as needed, and update the status of incidents as they progress from "new" to "in progress" and finally "resolved." This organized view helps coordinators distribute resources effectively and maintain oversight across multiple incidents.

Real-Time Communication:
CERA provides continuous updates to all parties. Residents receive notifications when their report is submitted, when a volunteer has been dispatched, and when the incident is resolved. Volunteers receive updates from coordinators about their assigned tasks. This real-time communication loop ensures transparency and minimizes uncertainty during emergencies.

Scalability and Future Features:
While the current project focuses on these core functions, CERA is designed with scalability in mind. Future enhancements may include live map tracking of volunteers, integration with official emergency

services, and offline reporting capabilities for areas with limited internet connectivity. These features highlight the potential of CERA to evolve into a comprehensive emergency management tool.

LITERATURE REVIEW

PREVIOUS WORK OR RESEARCH

Research in the field of mobile applications for emergency management shows a growing interest in using technology to support faster and more organized disaster response. A 2024 preprint on arXiv highlights the effectiveness of mobile-based platforms for disaster coordination, particularly in areas such as real-time updates, volunteer allocation, and resource optimization. Similarly, a study published in the Journal of Engineering Science and Technology in 2021 examined disaster management apps that focused on improving communication between citizens and authorities. A more recent review published in JETIR in 2023 analyzed crowd-sourced reporting and volunteer dispatch systems, concluding that such approaches have significant potential to improve emergency response times.

Despite these advances, current tools still exhibit gaps that limit their effectiveness. Most existing apps adopt a top-down approach, driven by government or agency priorities, rather than empowering local communities to self-organize. In addition, very few apps combine the essential features of incident reporting, volunteer coordination, real-time mapping, and offline support in a single platform. Our project addresses these limitations by developing a mobile-first solution that is community-centered, integrates essential features, and is designed for ease of use during high-stress situations.

ASSUMPTIONS, HYPOTHESES, AND BENEFITS

The development of CERA is based on several key assumptions. We assume that residents will adopt a mobile-first emergency solution if it is simple, reliable, and user-friendly. We also assume that volunteers can be effectively engaged through location-based notifications, and that peer testing during the development phase can provide valuable feedback even without real-world emergencies.

Based on these assumptions, our initial hypotheses are:

1. A mobile app can reduce the time it takes from reporting an incident to dispatching a volunteer.
2. A well-tested interface will improve trust and encourage more people to use the app.

The potential benefits of this project are significant. Communities will be able to respond to emergencies more quickly and in a more organized manner, reducing panic and confusion. Volunteers will have clear and specific tasks, replacing the scattered communication they often rely on today. Coordinators will benefit from structured oversight and real-time data, allowing them to make informed decisions. If successful, CERA has the potential to serve as a foundation for deployment in real-world communities, possibly in collaboration with local authorities and emergency services.

RESEARCH DESIGN

This project adopts an applied research design that emphasizes both the development and evaluation of a prototype mobile application, the Community Emergency Response App (CERA). Applied research is appropriate because the aim is not only to explore the concept of emergency coordination but also to deliver a working solution that addresses a real-world problem. The project is therefore practical in nature, focusing on implementation as well as assessment.

The research design is iterative, with features being introduced, tested, and refined in cycles. This ensures the app evolves through user interaction rather than assumptions. Each cycle involves developing a feature, testing it with users, and adjusting based on feedback. This approach is particularly valuable in emergency management, where usability, speed, and clarity are critical, allowing the team to address challenges early and deliver a functional, user-friendly prototype.

RESEARCH OBJECTIVES

The primary objective of this project is to design and implement a mobile application that supports community-based emergency reporting and volunteer coordination. Specifically, the project aims to test whether features such as real-time location tracking and push notifications can reduce the time between reporting an incident and dispatching a volunteer. Another objective is to evaluate the usability of the app through peer-based testing, ensuring that it remains simple and intuitive under stressful conditions. Finally, the project seeks to identify potential limitations and provide recommendations for future improvements and real-world deployment.

METHODOLOGY

OVERVIEW OF APPROACH

This project will follow an applied software development methodology with an iterative design process. The mobile application will be built using React Native to support both Android and iOS from a single codebase, while a RESTful backend with Express.js and MongoDB will handle data management and real-time updates. The system will incorporate role-based access to define clear functions for residents, volunteers, and coordinators.

The application will consist of several key modules. The first is an incident reporting feature that allows residents to log emergencies with photos, descriptions, and geolocation data. The second is a volunteer dispatch system that uses proximity-based notifications to connect available volunteers with nearby incidents. A live mapping module will display active incidents and volunteer locations, while an offline caching mechanism will provide limited functionality in low-connectivity areas. Finally, a coordinator dashboard will allow authorized users to approve volunteers, oversee incidents, and track their resolution.

JUSTIFICATION

React Native was selected as the development framework because it enables the use of a single codebase for both Android and iOS, making the process faster and more efficient. For the backend, Express.js and MongoDB provide a lightweight yet scalable solution that supports real-time updates and flexible data structures, which are essential for incident reporting and volunteer management. The choice of an iterative methodology is also intentional, as literature on emergency app development emphasizes the importance of rapid prototyping and feedback cycles to ensure usability and effectiveness. Finally, role-based access control is incorporated to guarantee both security and clarity of user functions, which is particularly critical in the context of emergency management systems where precision and reliability are vital.

DATA COLLECTION

Data for this project will be gathered through synthetic test incidents, user interaction logs, and simulated volunteer assignments to evaluate the app's functionality. Additionally, usability testing with peers or classmates will be conducted using wireframes and prototypes to collect qualitative feedback on the interface, workflow clarity, and overall user experience.

DATA ANALYSIS

The collected data will be examined using both quantitative and qualitative methods. Quantitative analysis will track metrics such as response times between incident creation and volunteer assignment, the number of successfully dispatched incidents, and offline synchronization success rates. Qualitative analysis will focus on usability, including peer feedback on interface intuitiveness, workflow efficiency, and overall user satisfaction. Together, these methods will provide a comprehensive evaluation of both system performance and user-friendliness.

TECHNOLOGIES USED

1. **Frontend (React Native)** – UI for Android/iOS with user roles (residents, volunteers, coordinators).
2. **Backend (Express.js)** – RESTful API for user, incident, and dispatch management.
3. **Database (MongoDB)** – Stores users, incidents, and activity logs.

4. **Authentication (JWT)** – Role-based access control.
5. **Geolocation & Maps** – Google Maps API for incident and volunteer tracking.
6. **Notifications** – Push notifications for incident updates.
7. **Offline Support (Low Priority)** – Local caching for reporting and viewing incidents without internet.
8. **Admin Dashboard (Optional)** – Web-based management panel for coordinators.

KEY STAKEHOLDERS

- Residents and community volunteers (end-users)
- Community coordinators or municipal staff
- NGOs and disaster relief organizations

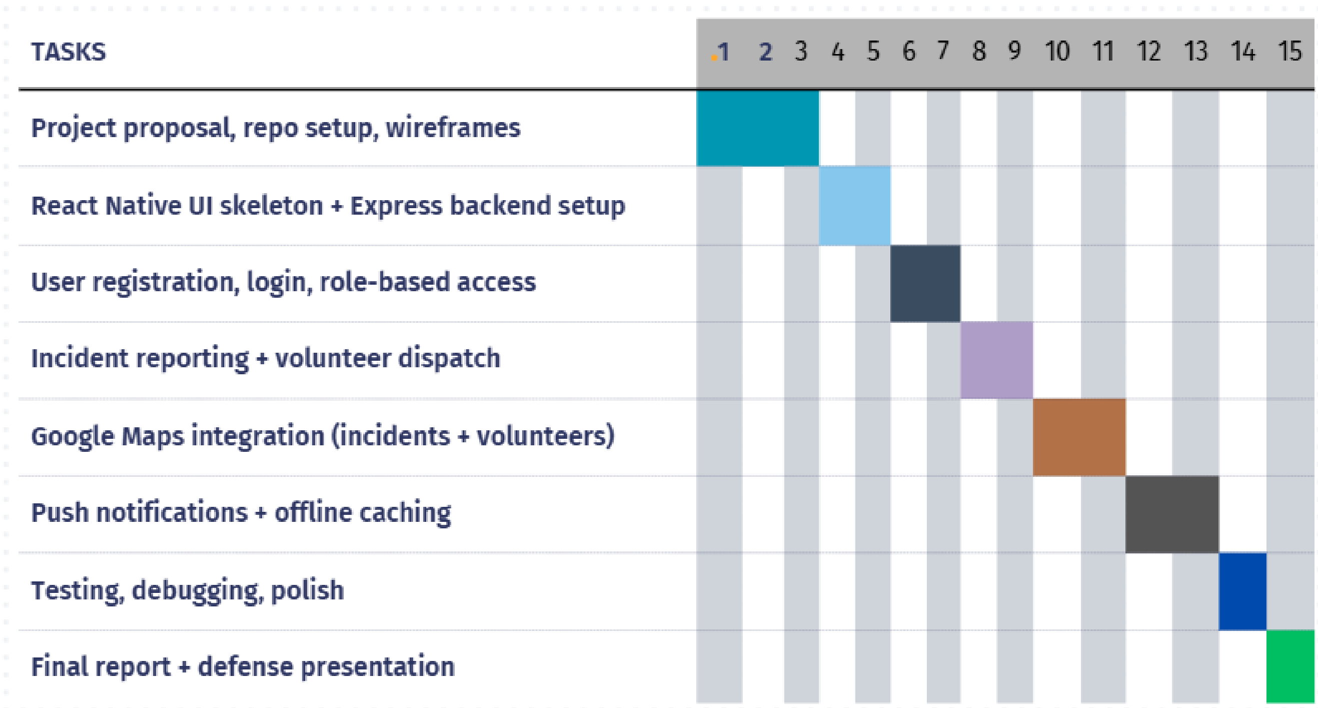
LIMITATIONS

- Testing relies on self created emergencies; real-world validation requires partnerships with municipalities.
- Backend hosted on basic or free version of cloud infrastructure may have performance limitations.

EXPECTED RESULTS AND IMPLICATIONS

We expect to develop a functioning mobile application that enables incident reporting, volunteer dispatch, and real-time mapping. The interface will be tested for usability to ensure clarity and ease of use for residents, volunteers, and coordinators. Documentation will capture implementation details, lessons learned, and recommendations for future enhancement and deployment. If successful, the prototype could serve as a foundation for real-world adoption, including integration with local authorities, NGOs, or other emergency management agencies.

PROJECT PLANNING AND TIMELINE



DETAILED IMPLEMENTATION OF GANTT CHART

WEEKS 1–3: PROJECT PREPARATION

User Stories

- As a project team, we need to define the scope, technical stack, and visualize the app so that we start with clarity and shared vision.

Tasks

- Create project proposal and requirements doc.
- Set up GitHub repo with correct naming conventions/folders.
- Build initial wireframes for mobile and admin interfaces.
- Assign user roles (resident, volunteer, coordinator/admin).

Deliverables

Project proposal, requirement documentation, GitHub repository setup, initial wireframes, assigned user roles.

WEEKS 4–5: BASIC APP & BACKEND SKELETON

User Stories

- As a user, I want to see a welcome and login/register screen so I can access the app securely.
- As a coordinator, I want the backend structure in place to start storing user and incident data.

Tasks

- Scaffold React Native project, show basic navigation and authentication screens.
- Setup Express.js backend with user and incident endpoints.
- Define MongoDB schemas for users, roles, incidents.
- Implement repository folder structure for code and docs.

Deliverables

Functional app skeleton, backend setup with user and incident endpoints, database schemas, basic navigation/authentication screens.

WEEKS 6–7: REGISTRATION, LOGIN, ROLE-BASED ACCESS

User Stories

- As a resident/volunteer, I want to register with my details and securely login, so I can use the app based on my role.
- As a coordinator, I want to approve or reject new volunteer registrations to ensure only qualified people join.

Tasks

- Develop registration and login screens (React Native).
- Implement backend authentication with JWT and password hashing (bcrypt).
- Role-based access/logins (restrict coordinator/admin views).
- Admin approval flow for volunteers.
- Security checkpoint: confirm password encryption, secure API routes, HTTPS setup (if deployed).

Deliverables

Fully functional registration and login screens, secure backend authentication, role-based access, volunteer approval workflow.

WEEKS 8–9: INCIDENT REPORTING & VOLUNTEER DISPATCH

User Stories

- As a resident, I want to quickly report an emergency, attach location and optional photo, so responders know what's happening.
- As a volunteer, I want to receive notifications about nearby incidents and accept or decline assignments.
- As a coordinator, I want to verify incidents before dispatch to reduce false reports.

Tasks

- Build report-incident screen (form, photo, geotag).
- Backend endpoint for new incident creation and volunteer dispatch logic.
- Volunteer notification logic (task list/accept decline).
- Incident filtering by type/status.
- Coordinator approval step for incidents before dispatch.

Deliverables

Incident reporting feature, volunteer dispatch notifications, coordinator approval system, incident filtering.

WEEKS 10–11: MAPS & GEOLOCATION INTEGRATION

User Stories

- As any user, I want to see incident locations and volunteer positions on a map, so I can visualize emergency activity in real time.
- As a coordinator, I want to assign volunteers based on region/proximity.
- As a coordinator, I want to see incident and volunteer stats, so I can monitor overall response levels.

Tasks

- Integrate Google Maps API in the mobile app.
- Show incidents and volunteers as map markers.
- Backend logic for location updates and proximity-based assignment.
- UI for map filters (by type, urgency, etc.).
- Basic analytics screen number of active incidents, volunteers available, completed responses.

Deliverables

Interactive map with incident and volunteer markers, proximity-based assignment logic, map filters, analytics dashboard.

WEEKS 12–13: PUSH NOTIFICATIONS & OFFLINE MODE

User Stories

- As a volunteer, I need push alerts when new emergencies are reported, so I can respond immediately.
- As a user, I need the app to function (view, report) even with limited or no connectivity.

Tasks

- Implement push notifications for new incidents/tasks.
- Local caching for offline support (incident list, reporting).
- Sync logic (store offline, send when online).

- UI for offline status alerts.
- Security/privacy review checkpoint: confirm safe storage of location data and user details.

Deliverables

Push notification system, offline functionality, secure local data storage, offline alerts.

WEEKS 14: TESTING, BUG FIXES, MIDTERM DEMO/VIDEO

User Stories

- As the project team, we need to verify all features, fix bugs, and demonstrate the core app functionality to stakeholders.

Tasks

- Write and run tests for major components (unit, integration, manual).
- Bug-tracking and resolution across modules.
- Prepare midterm demo and video showing core workflows.
- Seed test/demo data for realistic emergency scenarios (fire, flood, medical).

Deliverables

Tested and debugged app, midterm demo video, seeded test scenarios.

WEEKS 15: FINAL REPORT, UI POLISH, FINAL DEFENSE PREP

User Stories

- As a user, I want a polished, accessible, and user-friendly app so I can participate confidently during an emergency.
- As a coordinator/instructor, I want to complete documentation of the project—including lessons learned and future recommendations.

Tasks

- Finalize UI/UX (clean, accessible, onboarding/tips).

- Accessibility checks (contrast, readable fonts, simple workflows).
- Write final project report and supporting documentation.
- Prepare and practice for defense presentation.
- Update GitHub and ensure all deliverables are present.

Deliverables

Final polished app, accessibility-checked interface, complete project report, updated GitHub repository, prepared defense presentation.

TEAM RESPONSIBILITIES:

Dilpreet Sandhu: Backend development, database setup, authentication, push notifications

Harleen Kaur: Frontend development, UI/UX design, maps integration, testing

Both: Documentation, project planning, meetings

PROJECT AGREEMENT:

PROJECT DURATION: – 15 Sept 2025 To 27-Nov-2025

TEAM MEMBERS:

1. Dilpreet Sandhu
2. Harleen Kaur

1. AGREEMENT ON SCOPE OF WORK

We, the undersigned team members, agree to work collaboratively on the development of the **Community Emergency Response App (CERA)**. The project scope includes:

Scope of Work:

- Design and implement a mobile application for residents, volunteers, and coordinators.
- Modules to include:
  - Incident reporting with photos and geolocation
  - Volunteer dispatch with proximity notifications
  - Live mapping of incidents and volunteers
  - Push notifications and offline caching (low priority)
  - Admin dashboard for coordinators
- Backend development with Express.js and MongoDB
- Authentication with JWT for secure, role-based access.
- Testing, debugging, and usability evaluation.
- Final report, documentation, and defense presentation.

DELIVERABLES AND TIMELINE:

Week	Task / Milestone	Deliverable
1–3	Project proposal, repo setup, wireframes	Proposal + GitHub setup
4–5	React Native UI skeleton + Express backend setup	Initial frontend/backend code
6–7	User registration, login, role-based access	Auth module

Week	Task / Milestone	Deliverable
8–9	Incident reporting + volunteer dispatch	Reporting & dispatch module
10–11	Google Maps integration (incidents + volunteers)	Map view
12–13	Push notifications + offline caching	Notifications + offline support
14	Testing, debugging, polish	Midterm demo video
15	Final report + defense presentation	Final report + presentation

2. ROLES AND RESPONSIBILITIES

Team Members Agree To:

- Attend weekly meetings and participate in planning, discussion, and task completion.
- Complete assigned tasks on time and maintain code/documentation standards.
- Test individual components before integration.
- Update GitHub repository with all code and deliverables
- Communicate issues promptly to the team.

3. MEETING FREQUENCY AND FORMAT

- **Frequency:** Twice Every Week
- **Format:** Online via (Zoom, Google Meet) or in-person if necessary.
- **Agenda:**
    1. Review previous tasks and progress.
    2. Discuss challenges and blockers.
    3. Assign new tasks for the upcoming week.

4. COMMUNICATION AND TASK MANAGEMENT

- Team will use WhatsApp for daily updates and quick questions.
- Any changes to scope, timeline, or deliverables must be agreed upon by all members and communicated to the professor.

5. SIGNATURES

By signing this agreement, each team member confirms their understanding and acceptance of the project scope, responsibilities, and timelines.

Name	Signature	Date
Dilpreet Sandhu	Dilpreet Sandhu	15-Sept-2025
Harleen Kaur	Harleen Kaur	15-Sept-2025

This contract ensures **clarity, accountability, and smooth collaboration** for the successful completion of the CERA project.

WORK LOGS:

Dilpreet Sandhu:

Date	Task Completed	Hours Spent
4- Sept-2025	Created project group with Harleen. Explored different projects of Riipen and choose two of them 1.Glu 2.Weather Driver Decided to go with weather Driver after reading the description and discussed with group partner.	2 Hours
11-Sept-2025	Research more about the description of project to get the idea and explored many applications related to this topic.	1.5 Hours
13-Sept-2025	Had meeting on call with Harleen to know what exactly the requirements of projects. Juliet explained her about two projects but we both decide to go with none of them and instead of it we decided to create our own app from scratch. Explored various mobile and web applications for community services and volunteer coordination. Collected references and analyzed features to understand functionality and user interaction patterns.	2 Hours
14-Sept-2025	Initial research on emergency response apps. Added research notes for project proposal.	3 Hours

14-Sept-2025	Had a group meeting on call to discuss various project ideas and, after discussion, decided to focus on developing a Community Emergency Response App. Started Documentation Wrote Literature Review, Limitations , Expected Results and Implications, Create Gantt chart	2.5 Hours
15-Sept-2025	Wrote detailed implementation of Gantt chart with user stories	1.5 Hours

Harleen Kaur:

Date	Task Completed	Hours Spent
4- Sept-2025	Created Group with Dilpreet. Choose two projects of Riipen and decided to go with Weather Driver App as I wanted to create mobile app in react native.	2 Hours
7-Sept-2025	Explored the existing apps similar to weather driver app i.e. Drive Weather	1 Hour
13-Sept-2025	Had zoom meeting with Juliet and discussed about the projects.She offered two projects: 1.Customer Service Portal web app 2. Security Early Detection mobile app And then I researched on project. Discussed with Dilpreet about the projects and conclude to withdraw the request for ripen projects as the first project was out of scope for us as we want to create an app and first one was web app and second for large and complex for 2 members to complete within time limit. And decided to create our own app.	2.5 Hour
14-Sept-2025	Researched and brainstormed potential project ideas and discussed with Dilpreet and decided to go with his idea of CERA . Informed Juliet and Professor regarding the project withdraw.	3 Hours

	Started documentation Wrote Overview, methodology and data collection.	
15-Sept-2025	Wrote Scope of work and create project agreement	1.5 Hours

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